

VidVaani

Automated, low-cost dubbing of English technical lectures into Hindi

Nipun Batra

Computer Science & Engineering, IIT Gandhinagar

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The problem: technical education is locked in English

- India's best engineering lectures — NPTEL alone hosts **50,000+ hours** — are overwhelmingly in English.
- A large fraction of our students think and learn in Hindi and other Indian languages; a fast English lecture is a barrier, not a resource.
- **NEP 2020** explicitly calls for higher education content in Indian languages.
- Manual dubbing is slow and expensive: NPTEL's human-verified translation effort has taken years to cover a subset of courses in 11 languages.

Goal: given any lecture video, produce a natural-sounding Hindi version in minutes, for rupees — with the technical vocabulary students actually use.

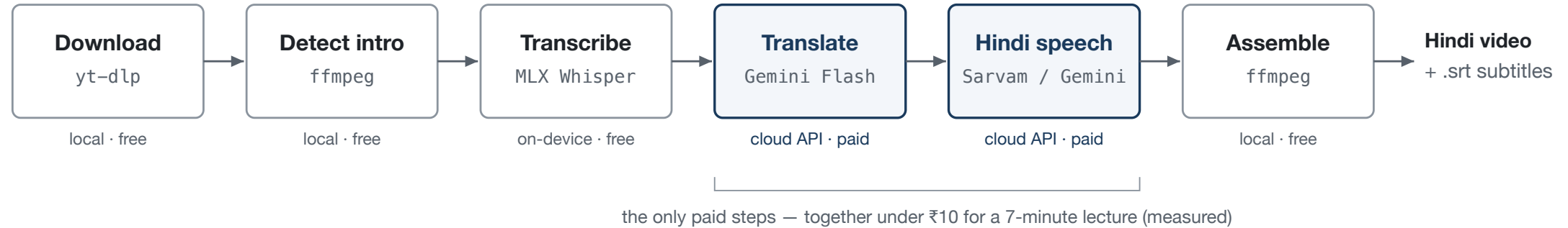
Existing options do not fit the lecture use case

Option	Cost	Why it falls short
YouTube auto-dubbing	Free	Only the channel owner can enable it; dubs are uneditable; no control over voice or terminology; widely criticised as robotic
Commercial dubbing SaaS (ElevenLabs, Rask, HeyGen, Dubverse, Murf)	₹3,000–11,000 per lecture	Priced for short marketing clips; monthly minute caps; each extra language billed again
Azure Video Translation	≈ ₹2,000 per lecture	Cheapest managed option, but limited voice choice and no terminology control
NPTEL / Bhashini human pipelines	Government-funded	High quality but slow, course-by-course; not self-serve

The gap: a self-serve tool an instructor can point at *any* video, with control over voice and vocabulary, at commodity API prices.

VidVaani: a six-stage open pipeline

Input: YouTube URL

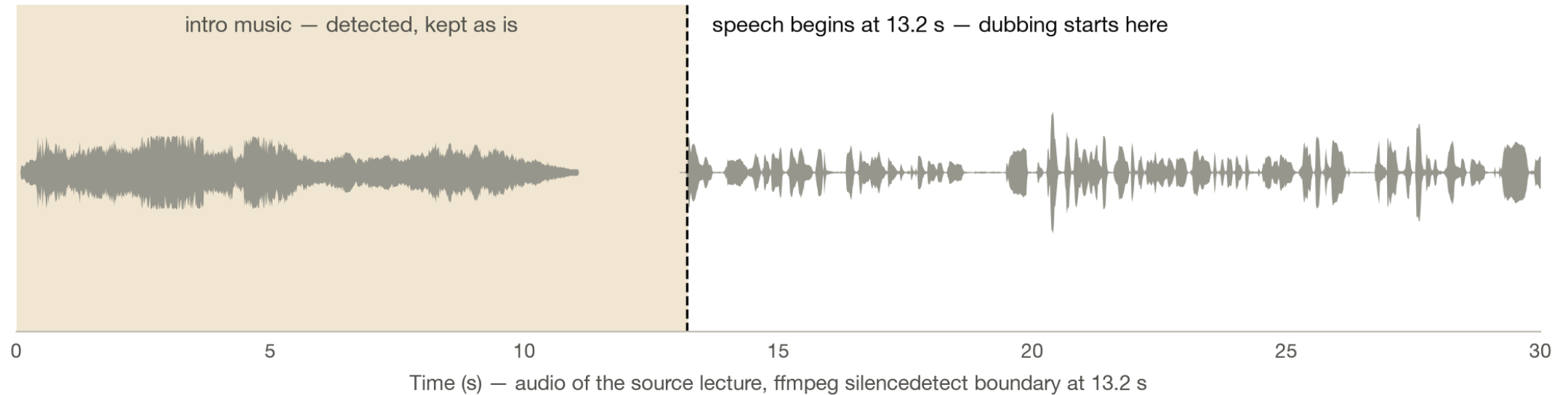


- Single command: `vidvaani dub <URL> --full -b sarvam -v abhilash`
- Transcription runs **on-device** (MLX Whisper on Apple Silicon) — audio never leaves the machine until translation.
- Every intermediate artifact is cached: re-dubbing with a new voice skips download, transcription, and translation.

The next five slides walk **one real lecture** — NPTEL CS7015 Lecture 1.1 (Prof. Mitesh Khapra, IIT Madras, 7 min) — through every stage.

Steps 1–2 — Download, and find where speech starts

`yt-dlp` fetches any YouTube URL or local file — no allowlist, no channel ownership needed. Then `ffmpeg` `silencedetect` locates the boundary between intro music and speech:



The music is **not dubbed over** — it is kept exactly as published, and dubbing begins on the first spoken word.

Step 3 — Transcribe on-device

MLX Whisper (`distil-large-v3`) runs locally on Apple Silicon — the audio never leaves the machine. Output: time-stamped English segments, grouped to ≤ 15 s:

Time	Whisper output (verbatim)
18.6 – 29.9 s	"In today's lecture is going to be a bit non-technical. We are not going to cover any technical concepts..."
29.9 – 41.1 s	"So, we hear the terms artificial neural networks, artificial neurons quite often these days..."
41.1 – 56.0 s	"And this history contains several spans across several fields, not just computer science..."

Transcribing the full 7-minute lecture took **17–20 s** (measured) — and it is free.

Step 4 — Translate for the clock, and the classroom

Constraints in the translation prompt: *speakable in the same duration; conversational (not literary) Hindi; technical terms stay in English.*

	Original (English)	VidVaani (Hindi)
13.2 s	"Hello everyone, welcome to lecture 1 of CS7015 which is the course on deep learning ."	"सभी को नमस्कार, CS7015 के लेक्चर एक में आपका स्वागत है, जो डीप लर्निंग का कोर्स है।"
29.9 s	"...we hear the terms artificial neural networks , artificial neurons quite often these days."	"...आजकल हम आर्टिफिशियल न्यूरल नेटवर्क्स, आर्टिफिशियल न्यूरॉन्स शब्द अक्सर सुनते हैं।"

Students say *gradient* and *neural network* — a dub that renders them as प्रवणता and तंत्रिका जाल (as fully-literal systems do) sounds foreign to the very audience it serves.

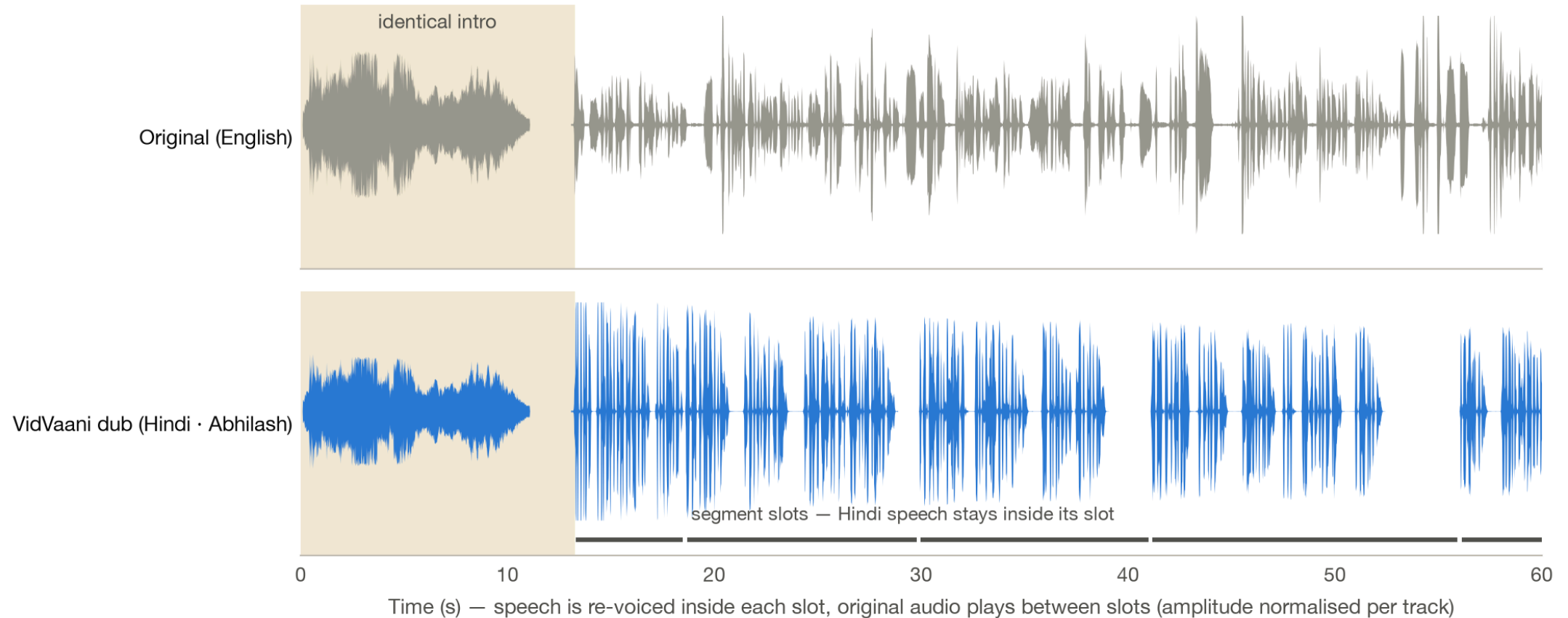
Step 5 — Synthesize, and fit the clock

Hindi runs ~15–35% longer than English. Per segment, following the automatic-dubbing literature:

1. The translator gets a **word budget** per slot (2.4 words/s, calibrated against the TTS voice's measured 2.8 words/s delivery) — fixing length at translation time beats stretching audio afterwards.
2. Synthesize at natural pace, trim trailing silence, measure with `ffprobe`.
3. If it fits ($\pm 5\%$) — done. Otherwise speed-adjust with an **asymmetric clamp (0.95x–1.35x)**: listeners tolerate faster speech far better than slowed speech.
4. Still too long? The clip **spills into the trailing pause** rather than being cut — professional dubs sacrifice timing, never content.
5. During pauses, the **original soundtrack plays** — atmosphere and intro music survive the dub.

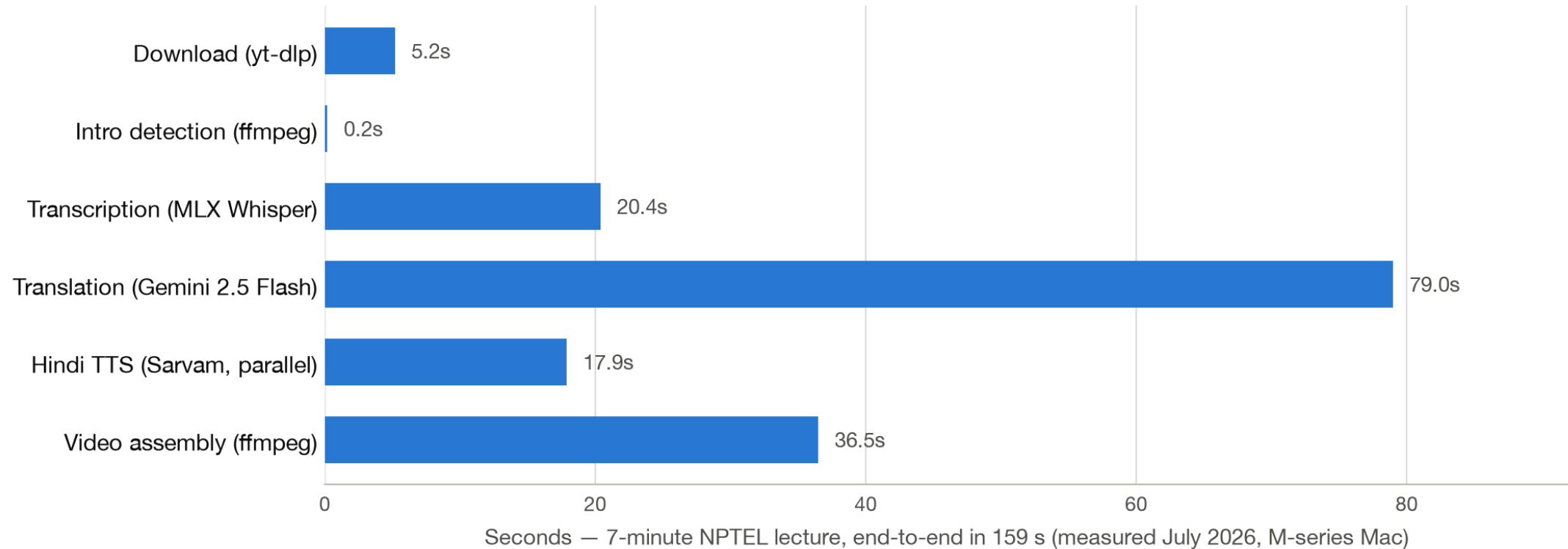
In the latest run of the example lecture, **all 33 segments fit their slots** — the word budget does most of the work before any audio adjustment is needed.

Step 6 — Reassemble: the result, seen in the signal



The intro is bit-identical to the source; every Hindi utterance starts on its original segment boundary; between utterances the original soundtrack plays. Plus a Hindi `.srt`, and optionally subtitles burned into the frame.

Measured performance — faster than real time



A 7-minute lecture dubs in 2½–4 minutes end-to-end on a laptop — the bottleneck is the translation API, not synthesis. Re-runs with a different voice take under a minute (cached transcript + translation).

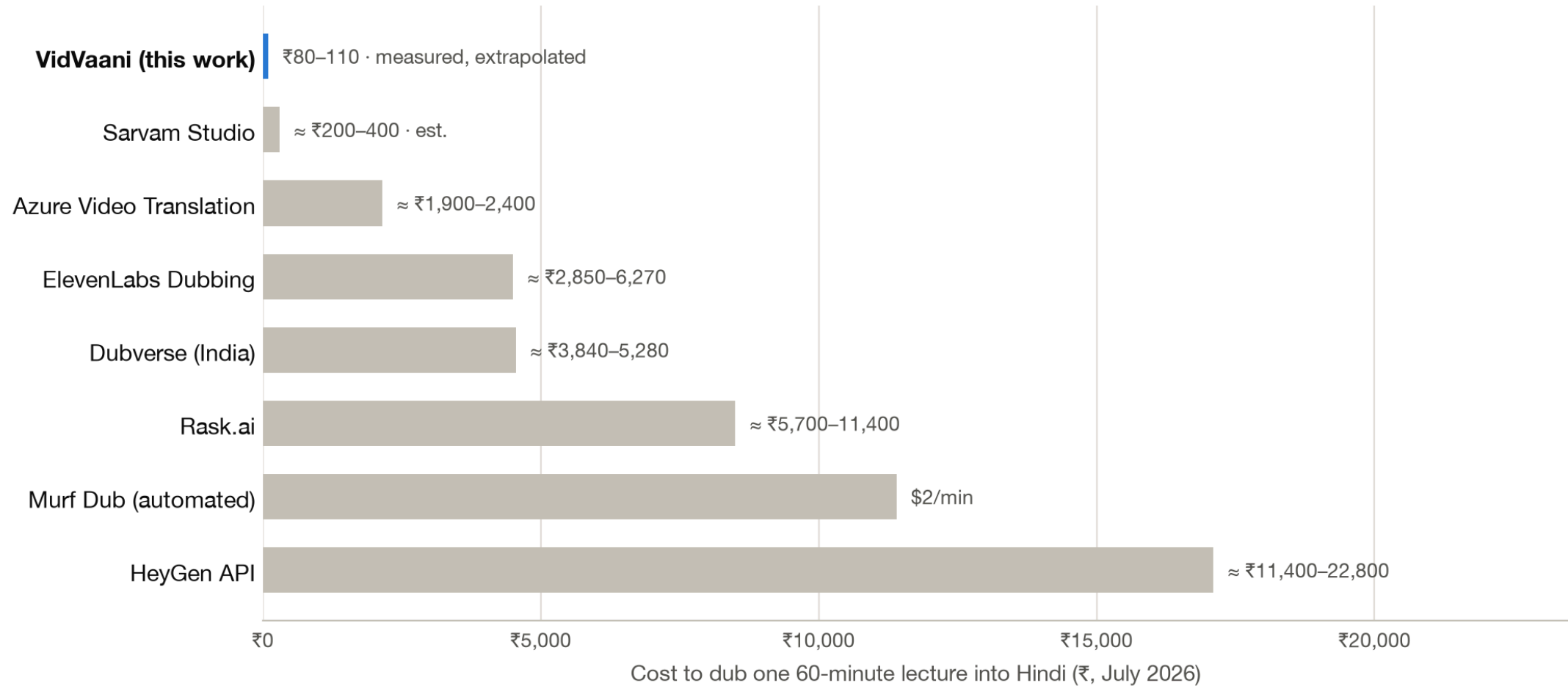
Measured cost — rupees, not lakhs

Measured on the same 7-minute lecture (5,307 Hindi characters):

	Translation	Hindi TTS	Total (7 min)	Extrapolated, 1 hour
Sarvam <code>bulbul:v2</code>	₹1.4	₹8.0	₹9.4 (\$0.10)	≈ ₹80
Gemini 2.5 Flash TTS	₹1.4	₹9.6	₹11.0 (\$0.12)	≈ ₹95
Edge TTS (free tier)	₹1.4	₹0	₹1.4	≈ ₹12

- A full **40-lecture course** ≈ **₹3,200–4,400** — less than one SaaS-dubbed lecture.
- Costs measured by the pipeline itself from API-reported token counts, at official July 2026 prices.
- Stable across runs: after calibrating the translation word budget, the same lecture re-ran at **₹7.9**.

Where this sits in the market



Midpoints of published prices, single target language, July 2026. YouTube auto-dubbing is free but channel-owner-only with no editorial control. Papercup/RWS (human-in-loop enterprise) quotes custom prices well above this scale.

Why not just use the platforms?

YouTube auto-dubbing

- Now rolled out to all channels — **but only the uploader** can enable it.
- A university cannot dub NPTEL or MIT content it does not own.
- Dubs cannot be edited — a mistranslated technical term can only be unpublished, not fixed.

Sarvam Studio & SaaS dubbers

- Produce formal, fully-translated Hindi (क्षेत्रफल for "area") — jarring for STEM learners used to Hinglish.
- No control over segment-level phrasing or terminology.
- VidVaani uses the **same Sarvam voices** with full editorial control, at API prices.

Notably, the popular open-source dubbing tools (pyvideotrans ★18k, VideoLingo ★17k) integrate **no Indian-language TTS at all** — VidVaani fills a real gap.

What a run looks like

```
$ vidvaani dub "https://youtube.com/watch?v=4TC5s_xNKSs" --full -b sarvam -v anushka
```

Downloaded: Deep Learning(CS7015): Lec 1.1 Biological Neuron

Transcribed: 33 segments

Generated subtitles: 4TC5s_xNKSs_hindi.srt

TTS: 33/33 segments (parallel)

Created: 4TC5s_xNKSs_hindi_anushka.mp4

Timing Breakdown

Download	5.2s	3.3%
Transcription	20.4s	12.8%
Translation	79.0s	49.6%
TTS Generation	17.9s	11.2%
Video Assembly	36.5s	22.9%
Total	159.2s	100%

API Costs

Translation	9,542 tokens	\$0.015
TTS (Sarvam)	5,307 chars	Rs 7.96
Total		Rs 9.4

Every run reports its own timing and cost — the numbers on the previous slides are this output, not estimates.

Demonstrations

nipunbatra.github.io/vidvaani — plays in any browser:

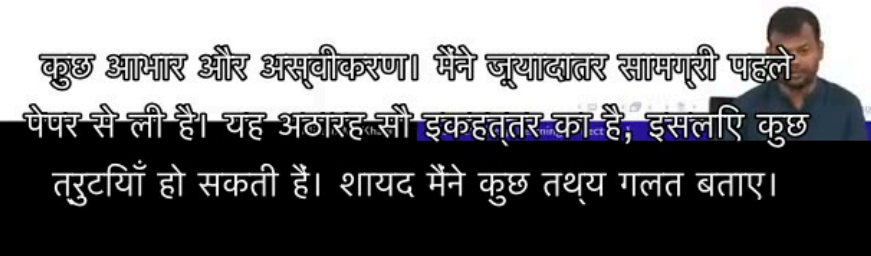
1. **58-second probability clip, ten ways** — original, Sarvam and Gemini voices (same translation, so the comparison is purely the voice), a **fully local** Gemma 4 + Qwen3-TTS version, the **lecturer's own cloned voice** (fine-tuned, 0.89 similarity), and Sarvam's own Dashboard output.
2. **Full 7-minute NPTEL lecture** — original beside Hindi dub; intro music preserved; five voices; burned-in subtitles.

What to listen for:

- Technical terms stay in English (*deep learning*, *CS7015*).
- Pauses land where the professor pauses.
- Native Indian prosody (Sarvam) vs. slight Western accent (Gemini).

Acknowledgements

- Most of this material is based on the article “Deep Learning in Neural Networks: An Overview” by J. Schmidhuber^[1]
- The errors, if any, are due to me and I apologize for them
- Feel free to contact me if you think certain portions need to be corrected (please provide appropriate references)



Dubbed lecture with generated Hindi subtitles burned in.



Scan to open the demo page.

Honest limitations

- **No human review loop yet** — NPTEL's own effort treats faculty verification as essential for pedagogy; ₹80/lecture buys the draft, not the sign-off.
- **Timing fit is still partly post-hoc** — word budgets plus a 0.95–1.35× speed clamp handle most segments, but very dense passages can still sound hurried (see next slide).
- Gemini TTS models are **preview** — pricing and behaviour may change; Edge TTS is an unofficial free endpoint, fine for prototyping only.
- No lip-sync. Voice cloning of the original lecturer now works locally (next slide) and stays gated on **consent** — the one published cloned demo is the author's own voice on his own lecture.
- Hindi first; Sarvam voices cover 11 Indian languages, so extension is natural.

Towards a fully local pipeline

Every cloud stage now has a working local, open-weights replacement — measured on the 58 s demo clip, Apple Silicon, July 2026:

Stage	Cloud (current)	Local (experimental)	Measured, local
Transcribe	— (already local)	MLX Whisper	17–20 s per 7-min lecture
Translate	Gemini 2.5 Flash	Gemma 4 31B (MLX 4-bit, Apache-2.0)	59 s for the clip (12B via ollama: 25 s)
Hindi speech	Sarvam API	Qwen3-TTS 1.7B (MLX, Apache-2.0)	1.6× real time , cloned (first run: ~4×)

- The "Fully local" demo card was produced this way — **₹0 per lecture, nothing leaves the machine**; at RTF 1.6 a 1-hour lecture $\approx$ a 2–2.5 h batch.
- **Voice cloning is now on the demo page**: the "Your own voice" card clones the author from 28 s of his English in the clip itself and speaks the Hindi dub in his voice — published with consent, since it is his own lecture (word-perfect STT round-trip, ~3 min end-to-end for the 58 s clip).
- Indic-specialised alternatives (AI4Bharat IndicF5, Chatterbox-hi) may beat Qwen3-TTS on Hindi naturalness; evaluation ongoing (see [docs/local-models.md](#)).

Cloning the lecturer's voice: five rungs in one day

Similarity = ECAPA cosine to his real voice. His own recordings score **0.93** against each other (the ceiling); other voices **~0.3**. Measured 12 Jul 2026.

Rung	Method	Compute	Similarity
1	Single take, cloned from a 28 s reference	laptop, 95 s	0.76
2	Best-of-16 search, scored + stitched	laptop, ~45 min	0.85
3	+ LoRA fine-tune on 30 min of his speech	laptop, 7.5 min training	0.86
4	+ official full SFT	lab A100, ~3 min training	0.89
5	+ 175-min verified dataset, 3-LR sweep	3 × A100	0.89 , best takes 0.83–0.87

Rungs 1–3 are a laptop; rungs 4–5 need the GPU tier — a $\approx$ ₹5 lakh workstation, or a lab server.

What moved the needle — and the guard-rail

In order of impact:

1. **Reference choice** — a clean mid-lecture window beat the clip's intro by +0.05 on its own.
2. **Fine-tuning on verified data** — every training clip ECAPA-checked against his voice; two candidate source videos failed verification outright and were rejected.
3. **Early stopping** — epoch 0 beat every later epoch, in every variant. More steps $\neq$ more you.
4. **Restoring the speaker encoder** — the official SFT silently drops those weights from checkpoints; reference cloning then runs on random weights until they are merged back.

The guard-rail: an STT round-trip on every published take. It caught the LoRA silently deduping a repeated clause ("x range 0–2 ... y range 0–2" losing the y), and the single highest-similarity take of the project being fluent gibberish in his voice. **Similarity alone is never a release criterion.**

Roadmap: smarter time alignment

We surveyed the automatic-dubbing literature (Amazon, Microsoft, IIT Madras, IWSLT) and re-based the alignment design on it — full notes in `docs/timing-alignment.md`:

Already implemented — word budgets in the translation prompt; asymmetric speed clamp; spill-into-pause instead of truncation; trailing-silence trimming.

Next:

1. **TTS-native pace instead of waveform stretching** — Sarvam exposes `pace` (0.3–3.0×); synthesis at the right speed beats resampling in every published comparison.
2. **Re-translate outliers once** — segments still needing $> 1.3\times$ speed get one "shorten by 25%" LLM pass; ~25% of En→Hi segments are expected to qualify.
3. **Sentence-boundary segmentation** with word-level timestamps ($\approx +4.5$ BLEU over blind packing).
4. **Terminology glossary** carried across translation batches for consistency.
5. **Evaluation study** — student MOS ratings and comprehension quizzes, Hindi dub vs. English original.

Summary

- **Any lecture → natural Hindi dub in minutes:** one command, six automated stages, technical vocabulary preserved.
- **Measured, not estimated:** a 7-min lecture in ~3 minutes for under ₹10; $\approx$ ₹80–110 per lecture-hour; a 40-lecture course for the price of a textbook.
- **20–100× cheaper** than commercial dubbing platforms, with editorial control none of them offer.
- **Runs fully offline if needed** — and can speak the dub in **the lecturer's own cloned voice** (0.89 similarity to his real voice, consent-gated, every take content-checked).
- Open source (MIT): github.com/nipunbatra/vidvaani
- Live demos: nipunbatra.github.io/vidvaani

Next: faculty-in-the-loop review, more Indian languages, and a student comprehension study.